23AD02: Artificial Intelligence

UNIT-I

COURSE OUTCOMES (COs):

At the end of the course, student will be able to

- CO1:Enumerate the history & Foundation of AI. (Understand – L2)
- CO2:Apply the searching algorithms for AI in problem solving.(Apply-L3)
- CO3:Choose the appropriate representation of knowledge. (Apply-L3)
- CO4:Choose the appropriate logic concepts. (Apply-L3)
- CO5:Understand the Expert Systems techniques in AI. (Understand –L2)

Text Books:

- Stuart Russell and Peter Norvig, “Artificial Intelligence: A Modern Approach” , 3rd Edition, Pearson
- Saroj Kaushik, “Artificial Intelligence”, Cengage Learning India, 2011
- Elaine Rich and Kevin Knight, “Artificial Intelligence”, Tata McGraw Hil

UNIT I -SYLLABUS

- **Introduction:** (TextBook-1 Chapter-1)
- What Is AI?
- The Foundations of Artificial Intelligence
- The History of Artificial Intelligence
- **Intelligent Agents:** (TextBook-1 Chapter-2)
- Agents and Environments
- The Concept of Rationality
- The Nature of Environments
- Structure of Agents
- Problem Solving Agents
- Problem Formulation

Some Definitions (I)

- The exciting new effort to make computers think … machines with minds, in the full literal sense.

  Haugeland, 1985

- The study of mental faculties through the use of computational models.

  Charniak and McDermott, 1985

Some Definitions (II)

- A field of study that seeks to explain and emulate intelligent behavior in terms of computational processes.

Schalkoff, 1990

- The study of how to make computers do things at which, at the moment, people are better.

Rich & Knight, 1991

What is Intelligence?

- Intelligence:
  - The capacity to learn and solve problems (Websters dictionary)
  - in particular,
    - the ability to solve novel problems
    - the ability to act rationally
    - the ability to act like humans
- Artificial Intelligence
  - build and understand intelligent entities or agents
  - 2 main approaches: "Engineering" versus "cognitive modeling".

What Is AI?

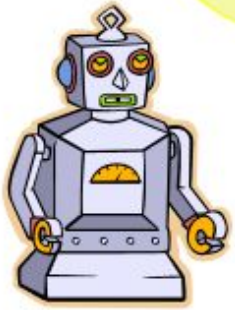
- **What is artificial intelligence?**
- It is the science and engineering of making intelligent machines, especially intelligent computer programs. It is related to the similar task of using computers to understand human intelligence, but AI does not have to confine itself to methods that are biologically observable.

- **Isn't there a solid definition of intelligence that doesn't depend on relating it to human intelligence?**
- Not yet. The problem is that we cannot yet characterize in general what kinds of computational procedures we want to call intelligent. We understand some of the mechanisms of intelligence and not others.

What's involved in Intelligence?

- Ability to interact with the real world
  - o to perceive, understand, and act
  - o e.g., speech recognition and understanding and synthesis
  - o e.g., image understanding
  - o e.g., ability to take actions, have an effect
- Reasoning and Planning
  - o modeling the external world, given input
  - o solving new problems, planning, and making decisions
  - o ability to deal with unexpected problems, uncertainties
- Learning and Adaptation
  - o we are continuously learning and adapting
  - o our internal models are always being "updated"
    - e.g., a baby learning to categorize and recognize animals

AI approaches

Systems that think like humans	Systems that think rationally
	
Systems that act like humans	**Systems that act rationally**
	

AI approaches

1. Acting humanly: The Turing Test approac

   The computer would need to possess the following capabilities.

   - natural language processing
   - Knowledge representation
   - Automated reasoning
   - Machine learning
   - (vision, robotics) for full test

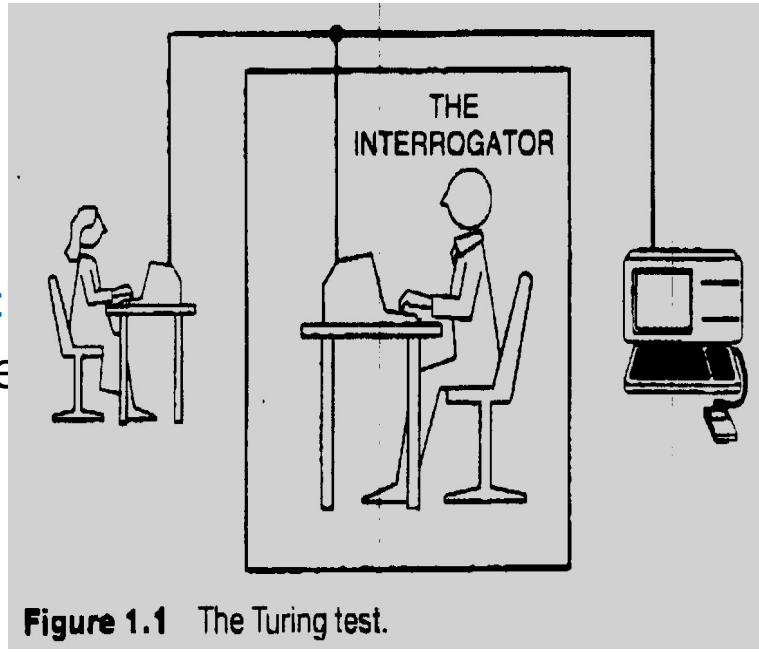


Figure 1.1 The Turing test.

- The Turing Test approach is based on the idea that a machine can be considered intelligent if it can convincingly pass a test that was proposed by British mathematician and computer scientist Alan Turing.
- The Turing Test involves a human evaluator engaging in a natural language conversation with a machine and a human, without knowing which is which. If the machine can successfully convince the evaluator that it is the human, then it is considered to have passed the Turing Test.

2. Thinking Humanly: **The cognitive modeling approach**

- How humans think?
- In 3 ways we can learn about human thoughts:
- 1. Introspection: trying to catch our own thoughts
- 2. Psychological experiments:
- 3. Brain Imaging:
- This approach emphasizes the importance of understanding human cognition and how it can be replicated in machines, rather than focusing solely on statistical or mathematical models.

3. Thinking Rationally: **The " laws of thought" approach**

- Logic
- Problems: how to represent and reason in a domain
- that is based on the principles of formal logic and reasoning. The laws of thought approach aims to develop AI systems that can reason logically and make decisions based on a set of predefined rules.
- This approach is often used in expert systems, where a knowledge base of facts and rules is used to solve complex problems in a particular domain.
- Ex: John is a human and all humans are mortal.

4. Acting Rationally: The Rational agent approach

– Agents: Perceive and act

- The rational agent approach is based on the idea of rationality, which involves making decisions that maximize the chances of achieving one's goals, given the available information and resources.
- The rational agent approach emphasizes the importance of designing agents that can reason under uncertainty and adapt to changing environments, rather than simply following a set of predefined rules.
- One example of the rational agent approach is reinforcement learning, which involves training an agent to make decisions in an environment based on rewards and punishments. The agent learns to maximize its rewards by trying different actions and observing the outcomes.

The Turing Test

(Can Machine think? A. M. Turing, 1950)



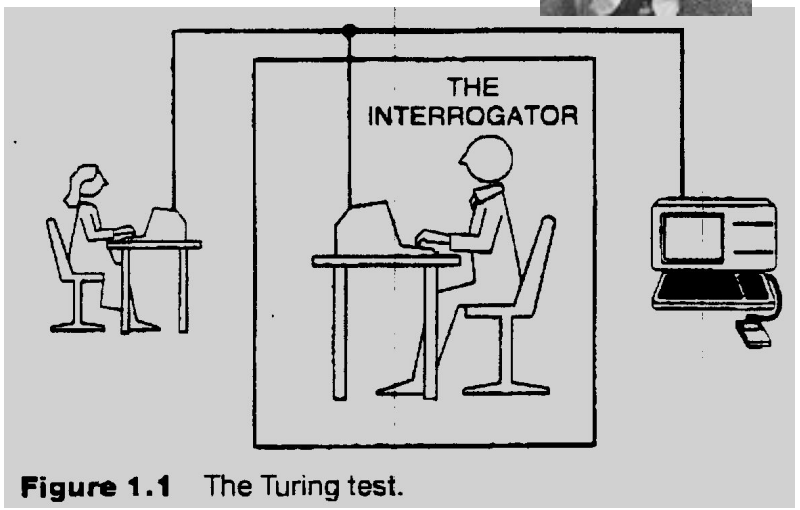


Figure 1.1 The Turing test.

- Requires
  - Natural language
  - Knowledge representation
  - Automated reasoning
  - Machine learning
  - (vision, robotics) for full test

History of AI

- McCulloch and Pitts (1943)
  - Neural networks that learn
- Minsky (1951)
  - Built a neural net computer
- Darmouth conference (1956):
  - McCarthy, Minsky, Newell, Simon met,
  - Logic theorist (LT)- proves a theorem in Principia Mathematica-Russel.
  - The name “Artficial Intelligence” was coined.
- 1952-1969
  - GPS- Newell and Simon
  - Geometry theorem prover - Gelernter (1959)
  - Samuel Checkers that learns (1952)
  - McCarthy - Lisp (1958), Advice Taker, Robinson’s resolution
  - Microworlds: Integration, block-worlds.
  - 1962- the perceptron convergence (Rosenblatt)

The Birthplace of "Artificial Intelligence", 1956

- Darmouth workshop, 1956: historical meeting of the precieved founders of AI met: John McCarthy, Marvin Minsky, Alan Newell, and Herbert Simon.

- A Proposal for the Dartmouth Summer Research Project on Artificial Intelligence. J. McCarthy, M. L. Minsky, N. Rochester, and C.E. Shannon. August 31, 1955. "We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it." And this marks the debut of the term "artificial intelligence."

- 50 anniversery of Darmouth workshop

History, continued

- 1966-1974 a dose of reality
  - Problems with computation
- 1969-1979 Knowledge-based systems
  - Weak vs. strong methods
  - Expert systems:
    - Dendral:Inferring molecular structures
    - Mycin: diagnosing blood infections
    - Prospector: recomending exploratory drilling (Duda).
  - Roger Shank: no syntax only semantics
- 1980-1988: AI becomes an industry
  - R1: Mcdermott, 1982, order configurations of computer systems
  - 1981: Fifth generation
- 1986-present: return to neural networks
- Recent event:
  - AI becomes a science: HMMs, planning, belief network

Abridged history of AI

- 1943 McCulloch & Pitts: Boolean circuit model of brain
- 1950 Turing's "Computing Machinery and Intelligence"
- 1956 Dartmouth meeting: "Artificial Intelligence" adopted
- 1952—69 Look, Ma, no hands!
- 1950s Early AI programs, including Samuel's checkers program, Newell & Simon's Logic Theorist, Gelernter's Geometry Engine
- 1965 Robinson's complete algorithm for logical reasoning
- 1966—73 AI discovers computational complexity
  Neural network research almost disappears
- 1969—79 Early development of knowledge-based systems
- 1980-- AI becomes an industry
- 1986-- Neural networks return to popularity
- 1987-- AI becomes a science
- 1995-- The emergence of intelligent agents

State of the art

- Deep Blue defeated the reigning world chess champion Garry Kasparov in 1997
- Proved a mathematical conjecture (Robbins conjecture) unsolved for decades
- No hands across America (driving autonomously 98% of the time from Pittsburgh to San Diego)
- During the 1991 Gulf War, US forces deployed an AI logistics planning and scheduling program that involved up to 50,000 vehicles, cargo, and people
- NASA's on-board autonomous planning program controlled the scheduling of operations for a spacecraft
- `Proverb` solves crossword puzzles better than most humans
- DARPA grand challenge 2003-2005, Robocup

Agents (chapter 2)

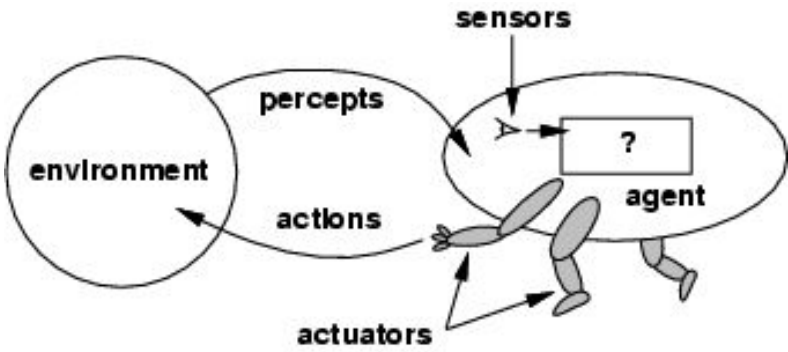
- Agents and environments
- Rationality
- PEAS (Performance measure, Environment, Actuators, Sensors)
- Environment types
- Agent types

Agents

- An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators

- Human agent: eyes, ears, and other organs for sensors; hands, legs, mouth, and other body parts for actuators
- Robotic agent: cameras and infrared range finders for sensors;
- various motors for actuators

Agents and environments

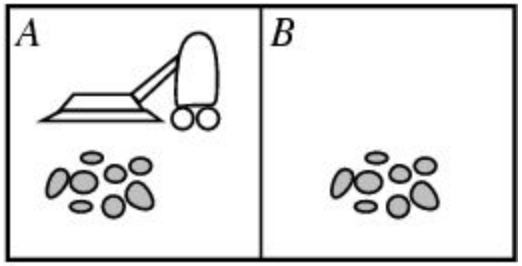


- The agent function maps from percept histories to actions:

$$[f: \mathcal{P}^* \to \mathcal{A}]$$

- The agent program runs on the physical architecture to produce *f*
- agent = architecture + program

Vacuum-cleaner world



- Percepts: location and contents, e.g., [A,Dirty]
- Actions: *Left*, *Right*, *Suck*, *NoOp*

Rational agents

- An agent should strive to "do the right thing", based on what it can perceive and the actions it can perform. The right action is the one that will cause the agent to be most successful
- Performance measure: An objective criterion for success of an agent's behavior

- E.g., performance measure of a vacuum-cleaner agent could be amount of dirt cleaned up, amount of time taken, amount of electricity consumed, amount of noise generated, etc.

Rationality

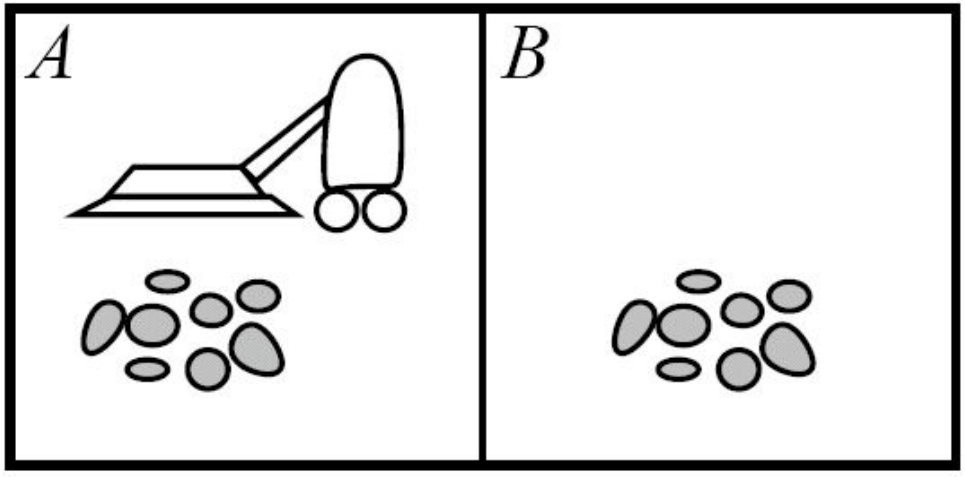
At any time it depends on four things:

- The performance measure that defines the criterion of success.
- The agents prior knowledge of the environment.
- The actions that the agent can perform.
- The agents percept sequence to date.

Rational agents

- Definition of Rational Agent: For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

Vacuum-cleaner world



Percepts: location and contents, e.g., $[A, Dirty]$

Actions: $Left$, $Right$, $Suck$, $NoOp$

Is this a Rational agent?

- Assume:
- The performance measure awards one point for each clean square at each time step, over a lifetime of 1000 time steps.
- The geography of the environment is known a priori but the dirt distribution and the initial location of the agent are not.
- The only available actions are Right, Left and Suck.
- The agent correctly perceives its location and whether that location contains dirt.

A vacuum-cleaner agent

Percept sequence	Action
$[A, Clean]$	$Right$
$[A, Dirty]$	$Suck$
$[B, Clean]$	$Left$
$[B, Dirty]$	$Suck$
$[A, Clean]$, $[A, Clean]$	$Right$
$[A, Clean]$, $[A, Dirty]$	$Suck$
⋮	⋮

function REFLEX-VACUUM-AGENT([*location*,*status*]) **returns** an action

if $status = Dirty$ **then return** $Suck$
else if $location = A$ **then return** $Right$
else if $location = B$ **then return** $Left$

What is the right function?

Can it be implemented in a small agent program?

The Nature of Environments

PEAS

To design a rational agent, we must specify the task environment

Consider, e.g., the task of designing an automated taxi:

Performance measure??

Environment??

Actuators??

Sensors??

PEAS

To design a rational agent, we must specify the task environment

Consider, e.g., the task of designing an automated taxi:

Performance measure?? safety, destination, profits, legality, comfort, ...

Environment?? US streets/freeways, traffic, pedestrians, weather, ...

Actuators?? steering, accelerator, brake, horn, speaker/display, ...

Sensors?? video, accelerometers, gauges, engine sensors, keyboard, GPS, ...

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits, minimize impact on other road users	Roads, other traffic, police, pedestrians, customers, weather	Steering, accelerator, brake, signal, horn, display, speech	Cameras, radar, speedometer, GPS, engine sensors, accelerometer, microphones, touchscreen

Figure 2.4 PEAS description of the task environment for an automated taxi driver.

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments	Touchscreen/voice entry of symptoms and findings
Satellite image analysis system	Correct categorization of objects, terrain	Orbiting satellite, downlink, weather	Display of scene categorization	High-resolution digital camera
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, tactile and joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, raw materials, operators	Valves, pumps, heaters, stirrers, displays	Temperature, pressure, flow, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, feedback, speech	Keyboard entry, voice

Figure 2.5 Examples of agent types and their PEAS descriptions.

Properties of task environments

- Fully observable vs. partially observable
- Deterministic vs. Stochastic Ex: chess, self driving cars
- Episodic vs. Sequential Ex: spot defective part, chess, taxi
- Static vs. Dynamic Ex: roller coaster, empty room
- Discrete vs. Continuous Ex: chess, self driving cars (no of moves)
- Single agent vs. multiagent

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

Figure 2.6 Examples of task environments and their characteristics.

Environment types

	Solitaire	Backgammon	Internet shopping	Taxi
Observable??				
Deterministic??				
Episodic??				
Static??				
Discrete??				
Single-agent??				

Environment types

	Solitaire	Backgammon	Internet shopping	Taxi
Observable??	Yes	Yes	No	No
Deterministic??				
Episodic??				
Static??				
Discrete??				
Single-agent??				

Environment types

	Solitaire	Backgammon	Internet shopping	Taxi
Observable??	Yes	Yes	No	No
Deterministic??	Yes	No	Partly	No
Episodic??				
Static??				
Discrete??				
Single-agent??				

Environment types

	Solitaire	Backgammon	Internet shopping	Taxi
Observable??	Yes	Yes	No	No
Deterministic??	Yes	No	Partly	No
Episodic??	No	No	No	No
Static??				
Discrete??				
Single-agent??				

Environment types

	Solitaire	Backgammon	Internet shopping	Taxi
Observable??	Yes	Yes	No	No
Deterministic??	Yes	No	Partly	No
Episodic??	No	No	No	No
Static??	Yes	Semi	Semi	No
Discrete??				
Single-agent??				

Environment types

	Solitaire	Backgammon	Internet shopping	Taxi
Observable??	Yes	Yes	No	No
Deterministic??	Yes	No	Partly	No
Episodic??	No	No	No	No
Static??	Yes	Semi	Semi	No
Discrete??	Yes	Yes	Yes	No
Single-agent??				

Environment types

	Solitaire	Backgammon	Internet shopping	Taxi
Observable??	Yes	Yes	No	No
Deterministic??	Yes	No	Partly	No
Episodic??	No	No	No	No
Static??	Yes	Semi	Semi	No
Discrete??	Yes	Yes	Yes	No
Single-agent??	Yes	No	No (except auctions)	No

The environment type largely determines the agent design

The real world is (of course) partially observable, stochastic, sequential, dynamic, continuous, multi-agent

The Structure of Agents

- The job of the agent is to design the AI program that implements the agent function(the mapping from percept to actions) The agent program runs on the physical architecture to produce *f*
- agent = architecture + program

```
function TABLE-DRIVEN-AGENT(percept) returns an action
  persistent: percepts, a sequence, initially empty
              table, a table of actions, indexed by percept sequences, initially fully specified

  append percept to the end of percepts
  action ← LOOKUP(percepts, table)
  return action
```

Figure 2.7 The TABLE-DRIVEN-AGENT program is invoked for each new percept and returns an action each time. It retains the complete percept sequence in memory.

A vacuum-cleaner agent

Percept sequence	Action
$[A, Clean]$	$Right$
$[A, Dirty]$	$Suck$
$[B, Clean]$	$Left$
$[B, Dirty]$	$Suck$
$[A, Clean]$, $[A, Clean]$	$Right$
$[A, Clean]$, $[A, Dirty]$	$Suck$
⋮	⋮

function REFLEX-VACUUM-AGENT([*location*,*status*]) **returns** an action

if *status* = *Dirty* **then return** *Suck*
else if *location* = *A* **then return** *Right*
else if *location* = *B* **then return** *Left*

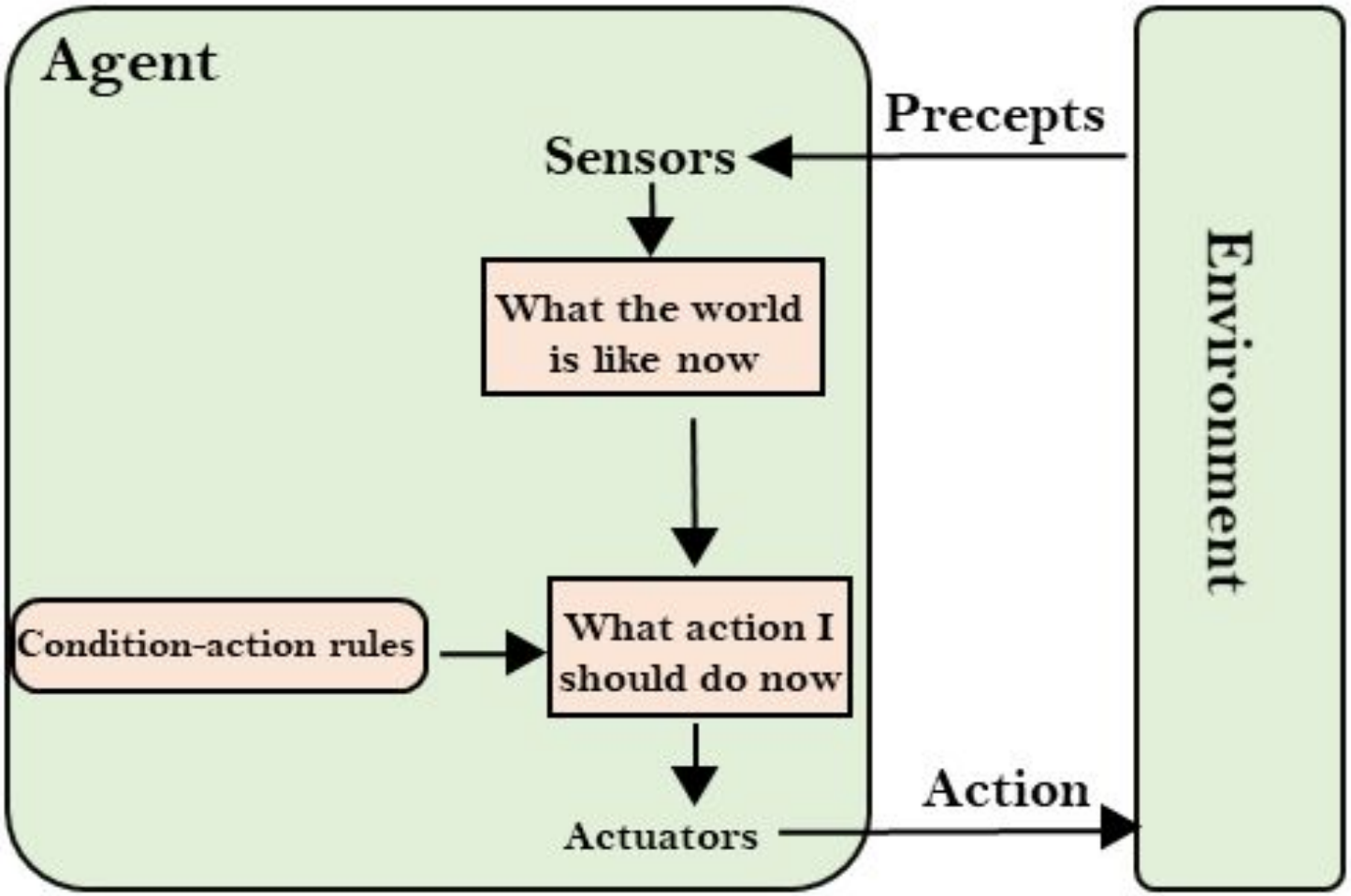
What is the right function?
Can it be implemented in a small agent program?

Agent Types

- Simple Reflex Agents
- Model-Based Reflex Agents
- Goal-Based Agents
- Utility-Based Agents

All these can be turned into Learning Agent

Simple Reflex agent:



Simple Reflex agent: (Cont..)

- The Simple reflex agents are the simplest agents. These agents take decisions on the basis of the current percepts and ignore the rest of the percept history.
- These agents only succeed in the fully observable environment.
- The Simple reflex agent works on Condition-action rule, which means it maps the current state to action. Such as a Room Cleaner agent, it works only if there is dirt in the room.

```
function SIMPLE-REFLEX-AGENT(percept) returns an action
  persistent: rules, a set of condition–action rules

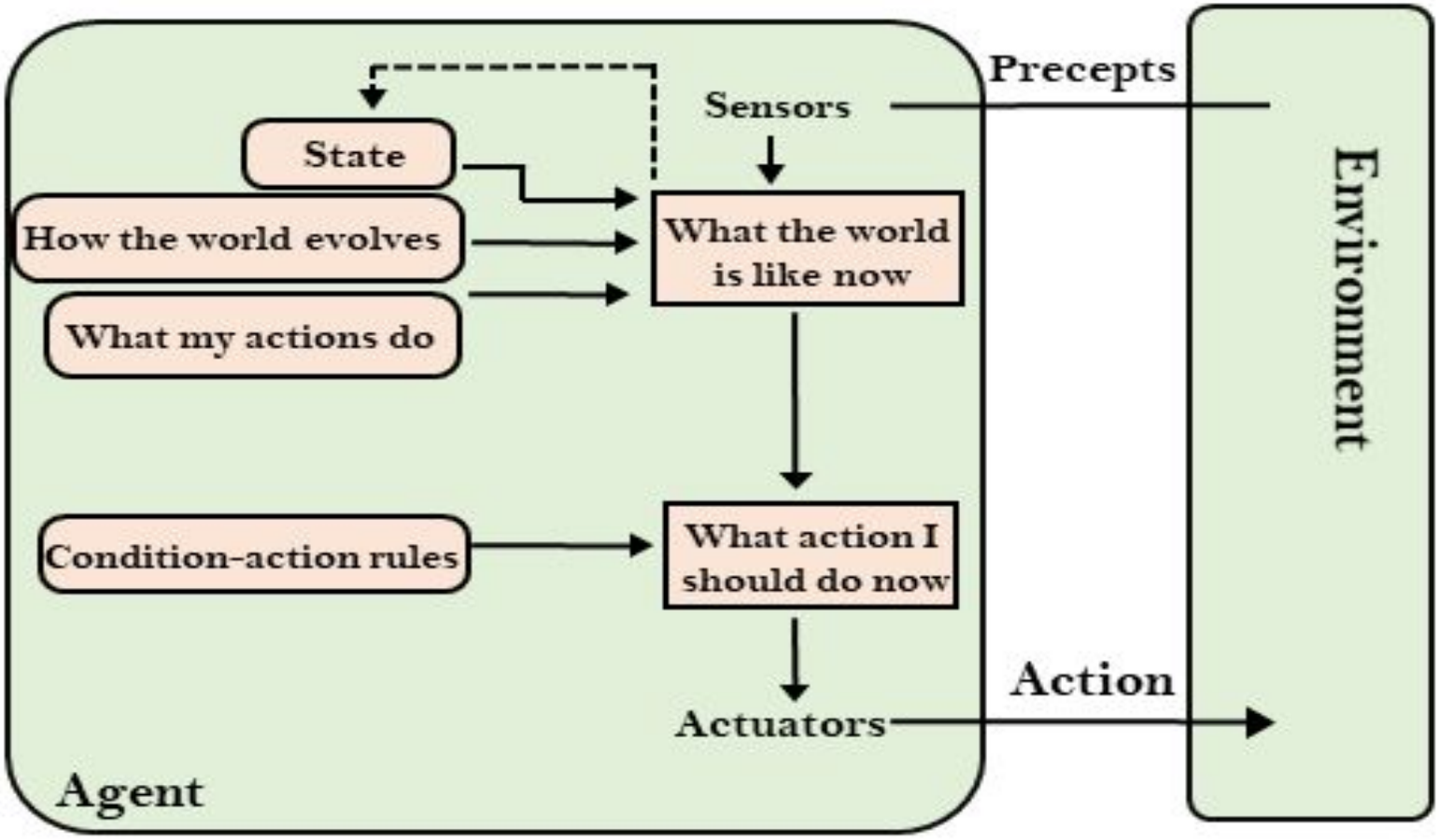
  state ← INTERPRET-INPUT(percept)
  rule ← RULE-MATCH(state, rules)
  action ← rule.ACTION
  return action
```

Figure 2.10 A simple reflex agent. It acts according to a rule whose condition matches the current state, as defined by the percept.

Problems with Simple reflex agents are :

- Problems with Simple reflex agents are :
- Very limited intelligence.
- No knowledge of non-perceptual parts of the state.
- Usually too big to generate and store.
- If there occurs any change in the environment, then the collection of rules need to be updated.

Model based Agent



Model based Agent (Cont..)

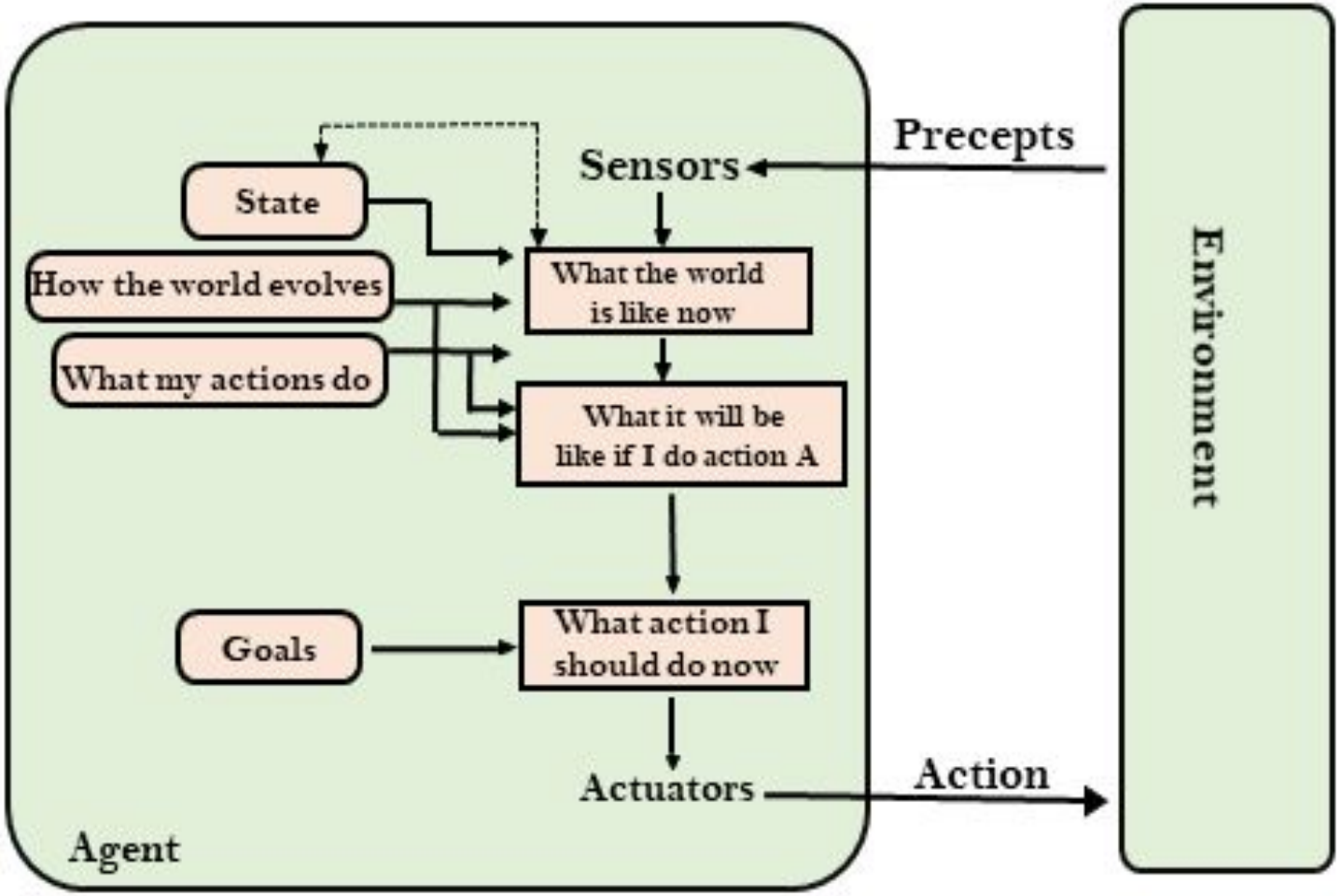
- The Model-based agent can work in a partially observable environment, and track the situation.
- A model-based agent has two important factors:
  - **Model:** It is knowledge about "how things happen in the world," so it is called a Model-based agent.
  - **Internal State:** It is a representation of the current state based on percept history.
- These agents have the model, "which is knowledge of the world" and based on the model they perform actions.
- Updating the agent state requires information about:
  - How the world evolves
  - How the agent's action affects the world.

```
function MODEL-BASED-REFLEX-AGENT(percept) returns an action
  persistent: state, the agent's current conception of the world state
              transition_model, a description of how the next state depends on
                    the current state and action
              sensor_model, a description of how the current world state is reflected
                    in the agent's percepts
              rules, a set of condition–action rules
              action, the most recent action, initially none

  state ← UPDATE-STATE(state, action, percept, transition_model, sensor_model)
  rule ← RULE-MATCH(state, rules)
  action ← rule.ACTION
  return action
```

Figure 2.12 A model-based reflex agent. It keeps track of the current state of the world, using an internal model. It then chooses an action in the same way as the reflex agent.

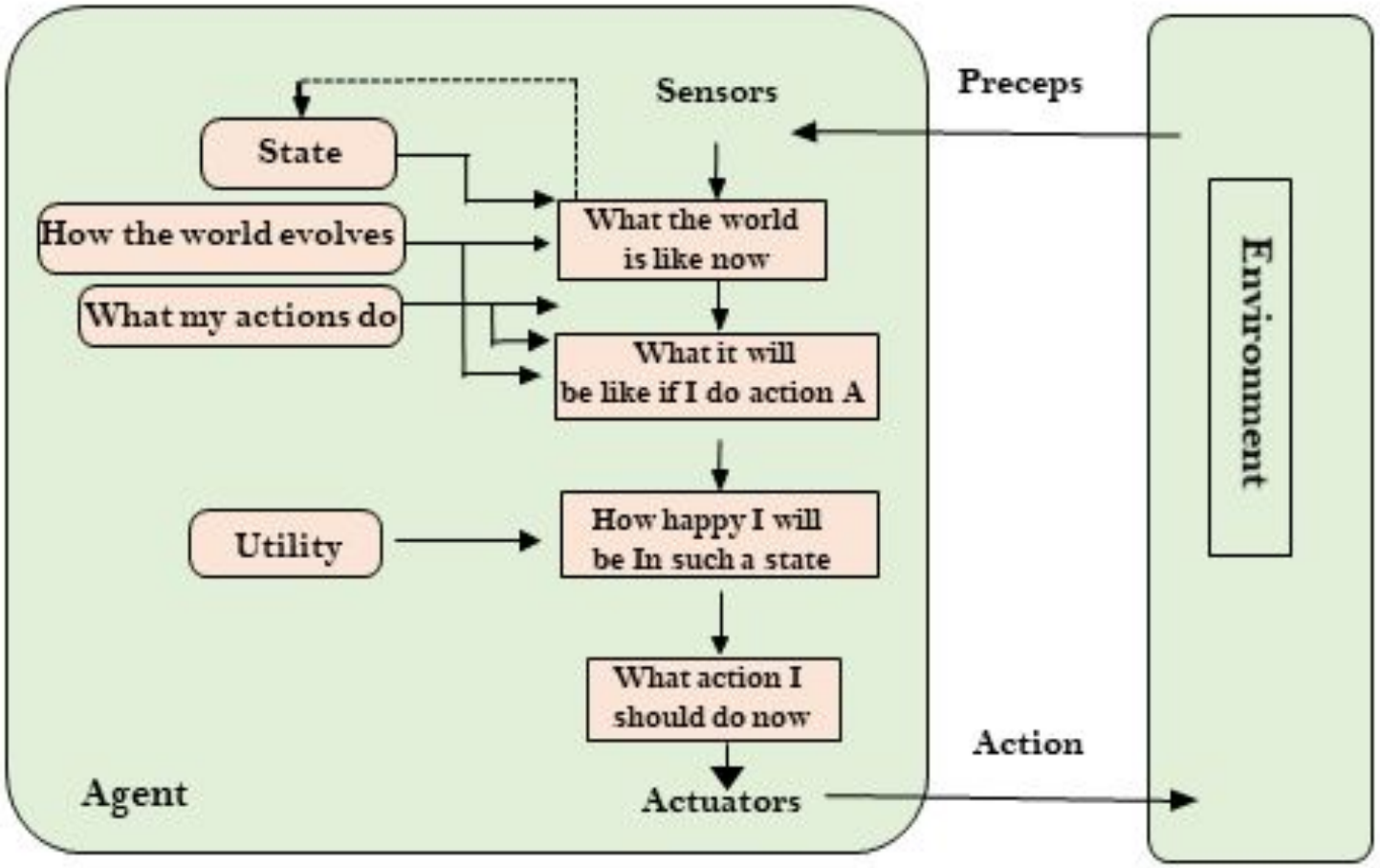
Goal based Agent



Goal based Agent (Cont..)

- The knowledge of the current state environment is not always sufficient to decide for an agent to what to do.
- The agent needs to know its goal which describes desirable situations.
- Goal-based agents expand the capabilities of the model-based agent by having the "goal" information.
- They choose an action, so that they can achieve the goal.
- These agents may have to consider a long sequence of possible actions before deciding whether the goal is achieved or not. Such considerations of different scenario are called searching and planning, which makes an agent proactive.

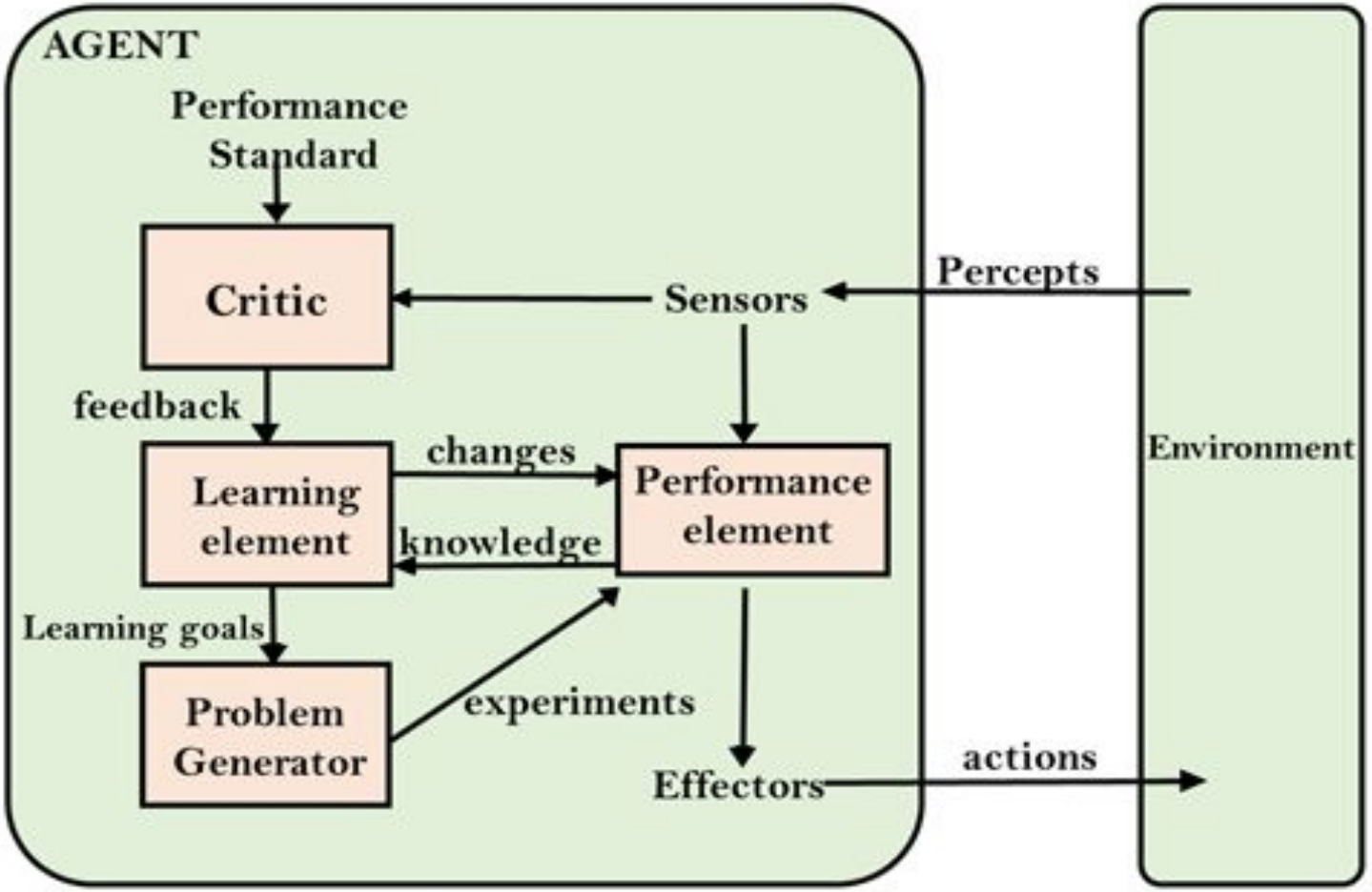
Utility-based agents



Utility-based agents (Cont..)

- These agents are similar to the goal-based agent but provide an extra component of utility measurement which makes them different by providing a measure of success at a given state.
- Utility-based agent act based not only goals but also the best way to achieve the goal.
- The Utility-based agent is useful when there are multiple possible alternatives, and an agent has to choose in order to perform the best action.
- The utility function maps each state to a real number to check how efficiently each action achieves the goals.

Learning Agents



Learning Agents

- A learning agent in AI is the type of agent which can learn from its past experiences, or it has learning capabilities.
- It starts to act with basic knowledge and then able to act and adapt automatically through learning.
- A learning agent has mainly four conceptual components, which are:
  - **Learning element:** It is responsible for making improvements by learning from environment
  - **Critic:** Learning element takes feedback from critic which describes that how well the agent is doing with respect to a fixed performance standard.
  - **Performance element:** It is responsible for selecting external action
  - **Problem generator:** This component is responsible for suggesting actions that will lead to new and informative experiences.
- Hence, learning agents are able to learn, analyze performance, and look for new ways to improve the performance.

Agent types

- Example: Taxi driver
- Simple reflex
  - **If** car-in-front-is-breaking **then** initiate-breaking
- Agents that keep track of the world
  - If car-in-front-is-breaking **then** initiate-breaking
  - needs internal state
- goal-based
  - If car-in-front-is-breaking **and** needs to get to hospital **then** go to adjacent lane and plan
  - search and planning
- utility-based
  - If car-in-front-is-breaking **and** needs to get to hospital alive **then** search of a way to get to the hospital that will make your passengers happy.
  - Needs utility function that map a state to a real function (am I happy?)

Summary

- What is Artificial Intelligence?
  - modeling humans thinking, acting, should think, should act.
- History of AI
- Intelligent agents
  - We want to build agents that act rationally

- Real-World Applications of AI
  - AI is alive and well in various “every day” applications
    - many products, systems, have AI components

Which of the following are AI and which are not. Choose yes, no, or “kind of” where kind of means that it both can be or can't be, depending on the viewpoint.

1. Spreadsheet that calculates sums and other pre-defined functions on given data

- The outcome is determined by the user-specified formula, no AI needed.

2. Predicting the stock market by fitting a curve to past data about stock prices

3. A GPS navigation system for finding the fastest route

4. A music recommendation system such as Spotify that suggests music based on the users' listening behavior

5. Big data storage solutions that can store huge amounts of data (such as images or video) and stream them to many users at the same time

- Photo editing features such as brightness and contrast in applications such as Photoshop

- Adjustments such as color balance, contrast, and so on, are neither adaptive nor autonomous, but the developers of the application may use some AI to automatically tune the filters. (kindof)

- Style transfer filters in applications such as Prisma that take a photo and transform it into different art styles (impressionist, cubist, ...)